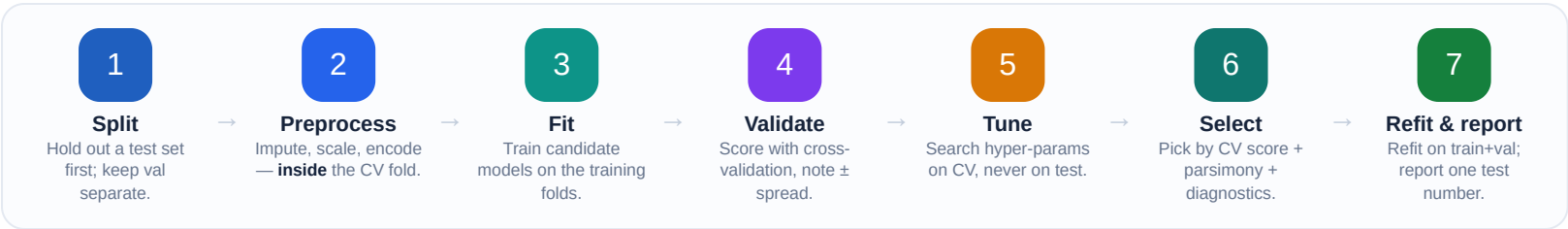


Build the Best Model: Workflow, Overfitting & Validation

The end-to-end path from raw data to a model you can trust — and the three numbers that tell you if it learned the signal or memorized the noise.

THE ONE RULE
The only score that estimates performance on brand-new data is cross-validated. Training and holdout are diagnostics — the CV mean ± spread is the truth.

THE END-TO-END MODELLING PIPELINE



🔒

Splitting & leakage guard

▶

Train / Validation / Test ≈ 60 / 20 / 20. Touch test **once**, at the very end.

▶

Stratify on the label for classification so each split keeps class balance.

▶

GroupKFold when rows share a source (same slab/patient/user) — split by the group.

▶

TimeSeriesSplit for temporal data — train on the past, test on the future only.

▶

Fit scalers, imputers, encoders, and feature selection **inside** each fold (use a Pipeline).

Leakage sources:

target leakage (a feature encodes the answer), train → test contamination (preprocessing fit on all data), and temporal leakage (future info in past rows).

🔑

The three R² — read them together

SCORE	WHAT IT IS	TRUST
Training R²	Fit on data the model has already seen.	Diagnostic only
Holdout R²	One untouched 20% split.	Quick check
Cross-val R²	Averaged over k held-out folds, with ± spread.	The headline

Overfit:

high train, big drop to CV, wide ±.

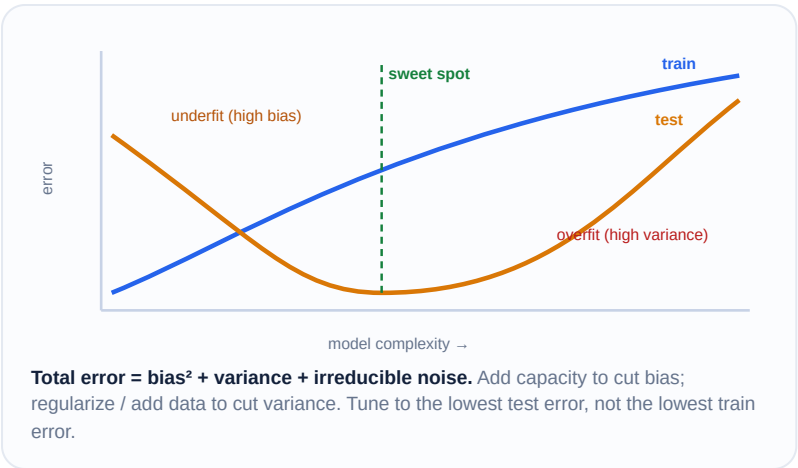
Underfit:

low everywhere.

Healthy:

all close & high, tight ±.

BIAS-VARIANCE & THE COMPLEXITY SWEET SPOT



Total error = bias² + variance + irreducible noise. Add capacity to cut bias; regularize / add data to cut variance. Tune to the lowest test error, not the lowest train error.

📈

Learning & validation curves

▶

Learning curve (score vs training-set size): gap that stays wide → variance, get more data / regularize. Both curves low & converged → bias, add capacity / features.

▶

Validation curve (score vs one hyper-parameter): find where CV peaks before it turns down.

▶

Early stopping: halt boosting / neural-net training when the validation score stops improving.

CROSS-VALIDATION SCHEMES — PICK THE ONE THAT MATCHES YOUR DATA

SCHEME	HOW IT WORKS	USE WHEN
k-Fold typically k = 5 or 10	Rotate each fold as the test set once; average the k scores.	Default for i.i.d. tabular data.
Stratified k-Fold	k-Fold that preserves class proportions in every fold.	Classification, especially imbalanced.
GroupKFold	No group appears in both train and test.	Repeated measures / clustered rows.
TimeSeriesSplit	Expanding window; always predict forward in time.	Any temporal / forecasting problem.
Repeated k-Fold	Run k-Fold several times with new shuffles; average.	Small data — tighten the estimate.
Nested CV outer + inner loops	Inner loop tunes hyper-params; outer loop measures the tuned model.	Honest score and tuning at once.

HYPER-PARAMETER SEARCH & CAPACITY CONTROL

🔧

Tuning strategies

▶

Grid search — exhaustive over a small, sensible grid.

▶

Random search — sample the space; wins when few params matter.

▶

Bayesian optimization (Optuna / Hyperopt) — model the score surface, spend trials where they help.

▶

Successive halving — start many configs cheaply, keep promoting the best.

Always tune against the CV score; keep the test set sealed.

⚙️

Regularization (shrink to fight variance)

Ridge (L2):

$\min \sum (y - \hat{y})^2 + \lambda \sum \beta_j^2$

— shrinks coefficients smoothly.

Lasso (L1):

$\min \sum (y - \hat{y})^2 + \lambda \sum |\beta_j|$

— drives some β to 0 (feature selection).

Elastic Net:

mixes L1 + L2 ($\lambda [\alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|^2]$).

Trees / NN:

limit `max_depth` , `min_samples_leaf` , learning rate, subsample, dropout, and use early stopping.

Regression Metrics: Meaning, Formula & Acceptance Range

Every number the target is continuous — what it measures, its equation, and the range that counts as good. Match the metric to the cost of the error and the target's scale.

ERROR METRICS — SIZE OF THE MISS

Metric	Formula	Meaning & When to Use	Target
MAE Mean Absolute Error	$(1/n) \sum y_i - \hat{y}_i $	Average miss in the target's own units; robust to outliers. Report to non-specialists.	lower; 0 = perfect
MSE Mean Squared Error	$(1/n) \sum (y_i - \hat{y}_i)^2$	Punishes big misses quadratically; the quantity most models optimize.	lower
RMSE Root Mean Squared Error	$\sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$	MSE back in target units; sensitive to large errors. Compare vs σ of y .	lower; < $\sigma(y)$
RMSLE Root Mean Sq. Log Error	$\sqrt{[(1/n) \sum (\ln(1+\hat{y}) - \ln(1+y))^2]}$	Penalizes relative error; good for skewed / multiplicative targets, under-prediction.	lower
MAPE Mean Abs. % Error	$(100/n) \sum (y_i - \hat{y}_i)/y_i $	Scale-free % error; fails if $y \approx 0$. Common in forecasting.	<10% good
sMAPE symmetric MAPE	$(100/n) \sum 2 \hat{y} - y / (y + \hat{y})$	Bounded, symmetric version; handles small y better.	lower
MedAE Median Absolute Error	$\text{median}(y_i - \hat{y}_i)$	Fully outlier-robust typical miss.	lower
Huber Huber loss (δ)	$\frac{1}{2}a^2 \text{ if } a \leq \delta; \delta(a - \frac{1}{2}\delta) \text{ else}$	Quadratic near 0, linear in the tails — MSE accuracy with MAE robustness.	lower

GOODNESS-OF-FIT — SHARE OF VARIANCE EXPLAINED

Metric	Formula	Meaning	Range
R² Coefficient of Determination	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Fraction of variance explained. Rises with any added predictor — never judge alone.	$(-\infty, 1]$
Adjusted R²	$1 - \frac{(1 - R^2) \cdot (n - 1)}{(n - p - 1)}$	R ² penalized for predictor count; use to compare models of different size.	$\leq R^2$
Explained Var.	$1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)}$	Like R ² but ignores systematic bias in the residuals.	$(-\infty, 1]$

Acceptance is field-dependent: physical/engineering fits often expect $R^2 > 0.9$; noisy human-behaviour or social data may be strong at $R^2 \approx 0.3$. Always compare against a baseline model.

MODEL SELECTION & PARSIMONY — FIT VS COMPLEXITY

Criterion	Formula	Rule
AIC Akaike Information Criterion	$2k - 2 \ln(\hat{L})$	Lower is better. k = #parameters, $\hat{L}$ = max likelihood. $\Delta\text{AIC} > 10$ = decisive.
BIC Bayesian Information Criterion	$k \cdot \ln(n) - 2 \ln(\hat{L})$	Heavier complexity penalty than AIC → favours simpler models on big n .
Mallows' Cp	$\text{RSS}_p / \sigma^2 - n + 2p$	Good subset when $C_p \approx p$. RSS = residual sum of squares.
F-statistic	$(\text{explained}/p) / (\text{residual}/(n-p-1))$	Tests whether the model beats an intercept-only fit (overall significance).
Log-likelihood $\ln(\hat{L})$	$\sum \ln f(y_i \text{model})$	Higher = better fit; the raw input to AIC / BIC.

RESIDUAL DIAGNOSTICS — IS THE FIT EVEN TRUSTWORTHY?

Check	Formula / Test	Flag & Action
VIF Variance Inflation Factor	$\text{VIF}_j = 1 / (1 - R_j^2)$	> 5–10 → multicollinearity; drop/combine predictors or use ridge.
Condition number	$\sqrt{(\lambda_{\max} / \lambda_{\min})}$	> 30 → unstable coefficients; center/scale, remove redundancy.
Durbin–Watson	$\frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2}$	Range 0–4; ≈ 2 ideal. <1.5 or >2.5 → autocorrelation (order/robust SE).
Breusch–Pagan	χ^2 test on residual variance	p < 0.05 → heteroscedasticity; use robust SE or transform y .
Cook's distance	$D_i = (r_i^2/p) \cdot h_i / (1 - h_i)$	$D_i > 4/n$ → influential point; inspect, don't blindly delete.
Jarque–Bera normality of residuals	$(n/6)(S^2 + \frac{1}{4}(K-3)^2)$	Over-rejects on big n — judge by skew ≈ 0 , kurtosis ≈ 3 , and Q–Q plot.

Classification Metrics & Tuning — with Equations

Everything the target is a class. Accuracy lies on imbalance — pick the metric that matches the mistake that actually costs you, then tune the threshold to it.

THE CONFUSION MATRIX — EVERYTHING IS BUILT FROM FOUR COUNTS

Actual +

Predicted +

TP
hit

Predicted -

FN
miss

Actual -

FP
false alarm

TN
correct reject

🔍 Which error costs more?

- Miss is dangerous (disease, defect, fraud) → maximize **Recall**.
- False alarm is costly (spam block, audit) → maximize **Precision**.
- Both matter → **F1 / Fβ**; ranking quality → **ROC-AUC / PR-AUC**.
- Imbalanced → never accuracy; use **PR-AUC, MCC, balanced accuracy**.

THRESHOLD METRICS — FROM THE FOUR COUNTS

METRIC	FORMULA	MEANS
Accuracy	$(TP+TN) / (TP+TN+FP+FN)$	Share correct. Misleads badly on imbalance.
Precision Positive Predictive Value	$TP / (TP + FP)$	Of flagged, how many were real.
Recall Sensitivity · TPR	$TP / (TP + FN)$	Of real positives, how many caught.
Specificity TNR	$TN / (TN + FP)$	Of real negatives, how many cleared.
FPR False Positive Rate	$FP / (FP + TN) = 1 - \text{Specificity}$	False-alarm rate (x-axis of ROC).
F1-score	$2 \cdot (P \cdot R) / (P + R)$	Harmonic mean of precision & recall.
Fβ-score	$(1+\beta^2) \cdot P \cdot R / (\beta^2 \cdot P + R)$	$\beta > 1$ favours recall; $\beta < 1$ favours precision.
Balanced Acc.	$(\text{Sensitivity} + \text{Specificity}) / 2$	Accuracy corrected for imbalance.
Jaccard IoU	$TP / (TP + FP + FN)$	Overlap of predicted & actual positives.

PROBABILITY METRICS — SCORE THE RANKING & CALIBRATION

METRIC	FORMULA / DEFINITION	RANGE · TARGET
ROC-AUC Area Under ROC Curve	$P(\text{score}(\text{pos}) > \text{score}(\text{neg}))$	0.5 chance · 0.7 fair · 0.8 good · 0.9+ excellent
PR-AUC Avg. Precision	$\sum (R_n - R_{n-1}) \cdot P_n$	Better than ROC on rare positives; baseline = prevalence.
Log loss Binary cross-entropy	$-(1/n) \sum [y \ln p + (1-y) \ln(1-p)]$	lower; 0 = perfect; punishes confident errors.
Brier score	$(1/n) \sum (p_i - y_i)^2$	lower; mean-squared probability error (calibration).
MCC Matthews Corr. Coef.	$(TP \cdot TN - FP \cdot FN) / \sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}$	[-1, 1]; balanced even on skew.
Cohen's κ kappa	$(p_o - p_e) / (1 - p_e)$	Agreement above chance; >0.6 substantial.

📊 **Threshold & curves**

Default 0.5 is rarely optimal. Move the cut to trade precision ↔ recall; pick it by cost or by **Youden's J = TPR – FPR**. ROC plots TPR vs FPR; PR plots P vs R (use PR when positives are rare).

⚖️ **Imbalance**

- Class weights / `class_weight`
- Resample: **SMOTE**, under-sample
- Threshold moving
- Judge on PR-AUC, MCC

🔧 **Calibration & multiclass**

Calibrate probabilities (Platt / isotonic); check a reliability diagram + Brier. **Multiclass avg**: macro (unweighted mean), micro (pool counts), weighted (by support).

Predicting an interval or a percentile
→ quantile loss + coverage. No
ground-truth groups → internal
cluster indices.

Quantile & Clustering Evaluation — with Equations

Two jobs mean-based metrics get wrong: predicting a percentile (score the quantile, not the mean) and learning structure with no labels (validate the geometry).

QUANTILE · A PERCENTILE — PREDICTION INTERVALS & T-LOSS

METRIC	FORMULA / DEFINITION	MEANING · TARGET
Pinball loss quantile loss, L_τ	$\tau(y - \hat{y})$ if $y \geq \hat{y}$; $(\tau - 1)(y - \hat{y})$ else	Asymmetric loss for the τ -quantile; under-/over-prediction weighted $\tau : 1 - \tau$.
Quantile (τ)	$\tau = 0.5$ median · 0.9 P90 · 0.1 P10	At $\tau=0.9$ an under-prediction is penalized 9× an over-prediction.
Coverage PICP	$(1/n) \sum 1 \{ L_i \leq y_i \leq U_i \}$	Share of actuals inside the interval; target $\approx 1 - \alpha$.
MPIW Mean Prediction Interval Width	$(1/n) \sum (U_i - L_i)$	Sharpness — narrower is better at correct coverage.
Interval / Winkler score	width + $(2/\alpha) \cdot (\text{miss distance})$	Rewards narrow intervals, penalizes points outside — the combined score.
Pseudo-R^2 (D^2)	$1 - L_\tau(\text{model}) / L_\tau(\text{baseline})$	Quantile analogue of R^2 vs a constant-quantile baseline.

Also check: **quantile crossing** (higher τ must predict $\geq$ lower τ) and consider **conformal prediction** for distribution-free intervals with guaranteed coverage.

CLUSTERING — INTERNAL INDICES (NO LABELS AVAILABLE)

INDEX	FORMULA / DEFINITION	RANGE · BETTER
Silhouette	$s = (b - a) / \max(a, b)$	$[-1, 1]$ · higher — a = mean intra-dist, b = nearest-cluster dist.
Davies–Bouldin DBI	$(1/k) \sum \max_j \neq i (\sigma_i + \sigma_j) / d(c_i, c_j)$	≥ 0 · lower — compact, well-separated = small.
Calinski–Harabasz Variance Ratio Criterion	$[\text{tr}(B)/(k-1)] / [\text{tr}(W)/(n-k)]$	higher — between- vs within-cluster dispersion.
Dunn index	min inter-dist / max intra-diameter	higher — separation over compactness.
Inertia / WCSS Within-Cluster Sum of Squares	$\sum_i \min_c \ x_i - \mu_c\ ^2$	lower — the elbow input (falls with every added k).

1 2 3 4 Choosing k (number of clusters)

- ▶ **Elbow**: plot WCSS vs k , take the bend where returns flatten.
- ▶ **Silhouette**: pick the k with the highest average silhouette.
- ▶ **Gap statistic**: $\text{Gap}(k) = E^*[\ln W_k] - \ln W_k$ vs a random reference; largest gap wins.

🔑 External indices (labels known)

INDEX	RANGE
ARI Adjusted Rand Index	$[-1, 1]$; 0 $\approx$ random
NMI Normalized Mutual Info	$[0, 1]$
V-measure homogeneity + completeness	$[0, 1]$
Fowlkes–Mallows	$[0, 1]$

🌿 Validate the clustering itself

Stability: re-cluster on bootstraps / subsamples — real structure is reproducible. **Standardize** features first (distance is scale-sensitive), match the **distance metric** to the data, and confirm clusters are meaningful, not artifacts of k or scaling.

